Metals with Multiple Charges

CHEMISTRY REFERENCE

These metals form more than one ion. The Roman numeral in the name gives the charge.

Metal	Ion		Ion	
Copper	**Cu^{+}**	copper(I)	**Cu^{2+}**	copper(II)
Iron	**Fe^{2+}**	iron(II)	**Fe^{3+}**	iron(III)
Lead	**Pb^{2+}**	lead(II)	**Pb^{4+}**	lead(IV)
Tin	**Sn^{2+}**	tin(II)	**Sn^{4+}**	tin(IV)
Mercury	**Hg_2^{2+}**	mercury(I)	**Hg^{2+}**	mercury(II)
Cobalt	**Co^{2+}**	cobalt(II)	**Co^{3+}**	cobalt(III)
Nickel	**Ni^{2+}**	nickel(II)	**Ni^{3+}**	nickel(III)
Chromium	**Cr^{2+}**	chromium(II)	**Cr^{3+}**	chromium(III)
Manganese	**Mn^{2+}**	manganese(II)	**Mn^{3+}**	manganese(III)
Gold	**Au^{+}**	gold(I)	**Au^{3+}**	gold(III)

Transition metals with one charge

Silver	**Ag^{+}**	always this charge — no Roman numeral
Zinc	**Zn^{2+}**	always this charge — no Roman numeral
Cadmium	**Cd^{2+}**	always this charge — no Roman numeral